

Reinforcement Learning using EEG signals for Therapeutic Use of Music in Emotion Management

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The Hook: Why This Research Matters

Music as Therapy: A Gateway to Emotion Regulation

- **Global Mental Health Challenge:** Mood disorders and emotional dysregulation are prevalent global issues. Traditional therapies often require intensive clinical supervision.
- **The Power of Music:** Music has a profound, scientifically proven capability to modulate human emotions, acting directly on the limbic system to alter affective states.
- **Why EEG?:** Electroencephalography (EEG) provides an objective, real-time window into cognitive and emotional brain dynamics, surpassing subjective self-reports.
- **The Need for Personalization:** Human emotional responses to music are highly subjective. A one-size-fits-all approach is ineffective.
- **The RL Shift:** We introduce Reinforcement Learning (RL) to transition from static music recommendations to an *adaptive, personalized therapeutic intervention* that actively guides a user's emotional state toward a positive target.

Problem Statement & Research Question

The Problem:

Current affective brain-computer interfaces (aBCI) and music recommendation systems are largely reactive—they identify how a user feels. However, there is a critical gap in building *active, closed-loop systems* capable of autonomously driving a person from a negative state (e.g., anger) to a desired positive state (e.g., happiness) using dynamic stimuli selection.

Research Questions:

- 1 Can a Reinforcement Learning agent effectively learn to select a sequence of music clips that navigates a user's emotional state toward a specific target ('happy') within a 2D Valence-Arousal space?
- 2 How do different RL paradigms (Q-Learning vs. SARSA vs. Double Q-Learning) compare in solving this stochastic emotional navigation problem?
- 3 What is the impact of reward shaping on mitigating state stagnation and improving convergence rates across diverse subjects?

Previous Work & Context

- **Emotion Recognition (DEAP Dataset):** Prior studies heavily rely on the DEAP dataset for emotion recognition, using various physiological signals (EEG, peripheral). Most focus purely on classification rather than state transition.
- **Dutta et al. (2020):** Proposed an initial Q-Learning framework for music emotion regulation using a discretized 20×20 Valence-Arousal space. We use this as our foundational benchmark.
- **Limitations of Current RL in Therapy:**
 - High variance in human subjects makes standard RL algorithms prone to overestimation bias.
 - Lack of efficient exploration constraints leads to stagnation where an agent repeatedly chooses clips that effect no emotional change.
 - **Our Contribution:** We expand upon previous work by implementing and conducting a rigorous comparative analysis of four algorithms (Q-Learning, SARSA, Double Q-Learning, Double SARSA) while introducing an explicit reward shaping penalty to break out of stagnant emotional loops.

Methods & Experimental Setup

Dataset: DEAP

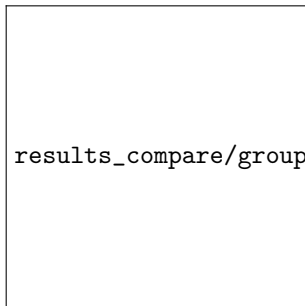
- 32 participants, 40 music video trials per subject (1 minute each).
- Features: EEG and physiological signals processed to map emotions.

Geneva Emotion Wheel & Valence-Arousal

- The continuous Valence-Arousal space is discretized into a 20×20 grid (400 states).
- 15 fine-grained emotions mapped into 8 main clusters: *happy, pleasant, exciting, calm, sad, negative, anger, fear*.

Evaluation Protocol

- **LOSO CV**: Leave-One-Subject-Out Cross Validation to ensure generalized inter-subject performance.
- Target Emotion: 'Happy' (Valence: ~ 0.905 , Arousal: ~ 0.425) defined as the goal state.
- Start state explicitly enforced to negative emotions (e.g.,



results_compare/grouped_gew.

Markov Decision Process (MDP) Formulation

- **State Space ($\mathcal{S}$):** A continuous Valence-Arousal space discretized into a 20×20 grid. Total of 400 possible discrete emotional states.
- **Action Space ($\mathcal{A}$):** Selection of localized music clips. Given the data and mapping, the agent chooses from up to 1240 available music-clip options. Each action aims to trigger an emotional state transition.
- **Transition Dynamics ($\mathcal{P}$):** Modeled entirely by the underlying EEG feature mapping of the human subjects. Specifically, the trajectory $s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_{target}$ depends on the empirical transitions found in the DEAP dataset.
- **Episode Structure:** Max length of 1000 step limits.
- **Grid Search:** Comprehensive grid search hyperparameter tuning over α (learning rate), γ (discount factor), and ϵ (exploration rate).

Reward Design & Shaping Mechanism

The problem of Stagnation: Often, an agent selects an action that fails to evoke an emotional response, resulting in $s_{next} = s_{current}$. Standard rewards do not sufficiently penalize this.

Reward Function Formulation:

$$R_{base}(s, a, s') = -\text{EuclideanDistance}(s', s_{target})$$

Base reward penalizes distance from the target emotion ('happy').

Reward Shaping Penalty (Φ):

$$\Phi(s, s') = \begin{cases} -100 & \text{if } s = s' \\ 0 & \text{otherwise} \end{cases}$$

This enforces a hard penalty for zero-transition actions, ensuring the agent continually explores the state space to find clips that induce actual emotional shifts.

Terminal Bonus: Goal reached yields an immediate +10 bonus to lock the policy.

Algorithm Focus: Q-Learning (Off-Policy)

Description: We replicated the Q-Learning framework by Dutta et al. (2020) for music emotion regulation. The agent operates on the discretized grid, estimating the absolute optimal future value regardless of the policy.

Update Rule with Reward Shaping:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') + \Phi(s, s') - Q(s, a) \right]$$

Performance Profile:

- **Why use it?:** It provides a strong baseline capable of finding the theoretical optimal path.
- **Results:** Strongest overall convergence with Reward Shaping (Exact: 30/32 subjects). Mean error 12.4°.
- **Drawbacks:** Susceptible to maximization bias in highly stochastic environments like human emotions, potentially overestimating the therapeutic value of certain music clips.

Algorithm Focus: SARSA (On-Policy)

Description: In contrast to off-policy Q-Learning, our SARSA agent updates its action-value function based on the actual trajectory followed by the exploratory ϵ -greedy policy.

Update Rule with Reward Shaping:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma Q(s', a') + \Phi(s, s') - Q(s, a) \right]$$

Performance Profile:

- **Why use it?:** It takes exploration into account during learning, making it a safer model if incorrect music recommendations are heavily penalized emotionally.
- **Results:** Suffered significant drop-off compared to Q-learning (Exact: 9/32). Mean angular error spiked to 77.2°.
- **Why it struggled?:** Emotion transitions are sparse and complex. On-policy exploration often led the agent into highly suboptimal loops ("sad", "negative" zones) from which it could not easily recover.

Algorithm Focus: Double Q-Learning

Description: Designed to mitigate the maximization bias inherent in standard Q-Learning—which often leads to the overestimation of music clip action values—we implemented Double Q-Learning (van Hasselt, 2010).

Update Rule with Reward Shaping: (Maintains tables Q^A and Q^B)

$$Q^A(s, a) \leftarrow Q^A(s, a) + \alpha \left[r + \gamma Q^B(s', \arg \max_a Q^A(s', a)) + \Phi(s, s') - Q^A(s, a) \right]$$

Performance Profile:

- **Why use it?:** It decouples action selection from action evaluation, providing a far more conservative, stable estimation of music-clip therapeutic rewards.
- **Results:** Extremely competitive (Exact: 29/32 with shaping). It successfully smoothed out overestimations while maintaining the exploratory effectiveness of off-policy learning.

Master Comparison Tables (32 Subject Benchmark)

Table: Performance **WITH** Reward Shaping Penalty

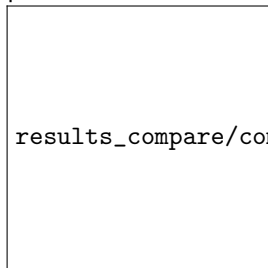
Algorithm	Exact Target	< 30° Err	< 60° Err	Mean Ang Error
Q-Learning	30/32	31/32	32/32	12.4° ± 8.9°
Double Q-Learning	29/32	25/32	30/32	24.1° ± 27.6°
SARSA	9/32	10/32	14/32	77.2° ± 56.1°
Paper Baseline (Q-Learning)	19/32	N/A	N/A	57.0° ± 2.8°

Table: Performance **WITHOUT** Reward Shaping

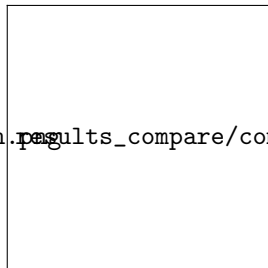
Algorithm	Exact Target	< 30° Err	< 60° Err	Mean Ang Error
Q-Learning	27/32	26/32	27/32	30.6° ± 46.7°
Double Q-Learning	24/32	22/32	25/32	34.2° ± 44.6°

Convergence Criteria & Visualizing Performance

Criteria: We evaluate the final angular error from the target 'happy' centroid. An exact arrival indicates optimal therapeutic outcome.



Angular Error Comparison

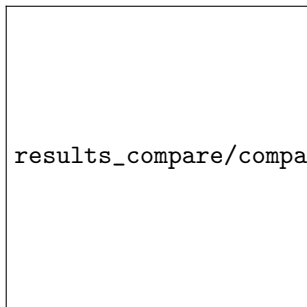


Convergence Bars (With Shaping)

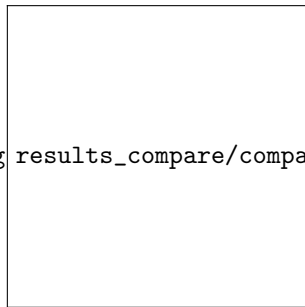
Observations:

- Across all metrics, Q-Learning coupled with reward shaping vastly outperformed the paper's baseline (12.4° vs 57.0°).
- Double Q-Learning provides closely trailing stability.
- Without shaping, mean angular errors doubled, proving the necessity of explicitly punishing stagnation.

The Direct Impact of Reward Shaping



Effect of Shaping on Path Metrics



Convergence Comparison (With vs Without)

Key Findings: The inclusion of the $\Phi(s, s')$ penalty completely shifts the agent's behavior from getting trapped in local negative minima to aggressively exploring the state space until a suitable music clip correctly transitions the emotion. Q-Learning jumped from 27 to 30 exact convergences.

Interpreting the Findings: Trajectory Deep Dive

Analyzing Specific Subject Emotional Paths

Case 1: SARSA failure mode (Subject S01 — Err=57.8°)

- *Path:* **anger** → **anger** → **pleasant** → **pleasant** → **pleasant** → **pleasant** → **pleasant**
- *Interpretation:* SARSA's on-policy updates caused the agent to settle for a locally safe, non-target emotion ("pleasant" rather than our target "happy"). It found a clip that kept the user stable and refused to explore further, missing the therapeutic mark.

Case 2: Double Q-Learning success (Subject S05 — Err=10.6°)

- *Path:* **anger** → **anger** → **fear** → **happy** → **happy** → **happy** → **happy**
- *Interpretation:* Double Q-learning optimally maneuvered the user state through an intermediate state (fear/high arousal transition), decoupled the value maximization, and correctly arrived and stabilized perfectly at the target "happy" state.

Future Directions: Increasing System Reliability

Integrating a Deep Sequential Emotion Classifier

While the RL agent handles policy mapping, we could drastically improve real-time reliability by integrating the underlying sequential raw EEG data through a deep classifier.

Proposed Bidirectional LSTM Architecture:

- **Input:** Sequence of 70 steps
- **Layers:** Bidirectional LSTM (256 units) $\rightarrow$ 3 stacked LSTMs (256 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 32)
- **Regularization:** Heavy Dropout integration between layers.
- **Output:** Dense layers classifying the 2D Valence-Arousal grid.
- **Parameters:** $\sim$ 786,610 trainable parameters.

Why add this? By integrating this sequential LSTM model alongside the RL agent, we can guarantee that the state inputs (s) fed into the RL mapping are robust against noise. It acts as an active feature-extractor validator for the agent.

Summary & Main Takeaways

Conclusions:

- 1 **RL for Emotion Regulation Works:** We successfully modeled emotional modulation via music as an MDP. The RL agents dynamically adapted sequence generation to pull subjective affective states out of negativity.
- 2 **Off-Policy superiority in Therapy:** Q-Learning and Double Q-Learning significantly outperformed SARSA. In highly noisy human environments, evaluating the absolute optimal path is more important than evaluating the exploratory policy path.
- 3 **Reward Shaping is Crucial:** Implementing a stagnation penalty completely eliminated zero-transition loops, boosting our Q-Learning convergence to 12.4° mean error, greatly improving upon the 57.0° baseline reported in prior literature.

Next Steps:

- Real-time user testing with live BCI hardware.
- Integrating our proposed deep Bidirectional LSTM classifier into the state formulation to increase reliability and noise resilience.

Thank You!

Questions?